
SUPPLEMENTARY EXERCISES

1 Probability basics

Exercise 1. We toss two coins and we consider the following events:

- | | |
|---------------------------------------|---------------------------------------|
| E1: obtain heads on the first coin | E6: obtain tails at least on one coin |
| E2: obtain tails on the first coin | E7: obtain heads and tails |
| E3: obtain heads on the second coin | E8: heads do not occur |
| E4: obtain tails on the second coin | E9: obtain two heads |
| E5: obtain heads at least on one coin | |

Construct the events: $E1 \cup E2$, $E1 \cap E3$, $E5 \cap E6$, $E7 \cup E5$, $E2 \cap E4$, $E5 \cup E9$.

Exercise 2. In a communication system, three messages are recorded. Consider the following events:

$A_i = \{\text{message } i \text{ is recorded correctly}\}$ $\bar{A}_i = \{\text{message } i \text{ is not recorded correctly}\}$

Express in function of A_i and $\bar{A}_i$ the following events:

- E_1 : all three messages are recorded correctly
 E_2 : none of the three messages is recorded correctly
 E_3 : at least one message is recorded correctly
 E_4 : at least one message is not recorded correctly
 E_5 : not less than two messages are recorded correctly
 E_6 : not more than one message is recorded correctly
 E_7 : the first message to be correctly recorded is the third one

Exercise 3. Consider A, B, C three events occurring during an experiment. Show that if A and B are incompatible then $A \cap C$ and $B \cap C$ are incompatible as well.

Exercise 4. Consider A and B two events. Show that $A \cup B$ can be written as the union of incompatible events.

Exercise 5. We consider the following experiments and events. Express the contrary events.

a) Experiment: In a communication system two messages are transmitted.

Event: $E = \{\text{the two messages are correctly recorded}\}$

b) Experiment: A ball is extracted from a box containing a white balls, b black balls and c red balls.

Event: $E = \{\text{the extracted ball is white}\}$.

c) Experiment: In a communication system five messages are transmitted.

Event: $E = \{\text{not less than three messages are correctly transmitted}\}$.

Exercise 6. Write the sample spaces of the following experiments: a) a coin is tossed three times; b) a coin and a die are thrown simultaneously; c) two distinct objects a and b are placed in two boxes A and B; d) a group of 25 students elect two representatives for a committee e) a coin is tossed until two heads or two tails are obtained successively; f) a polyhedron with k faces is rolled.

Exercise 7. Three dice are rolled. Express the events A: on the first and the third dice, the number 1 has been obtained and B: the sum of the numbers obtained on the three dice is less than 4. Find $A \cap B$.

Exercise 8. The experiment consists of randomly choosing an element of the set $A = \{a, b\}$ and an element of the set $B = \{a, b, c\}$. a) Write the elements of the set $A \times B$. b) What is the probability that the chosen elements are the same?

Exercise 9. A box contains 40 balls: 10 white, 25 black and 5 red balls. Three balls are extracted from this box. What is the probability that the three balls are of different colors?

Exercise 10. The numbers $1, 2, \dots, n$ are placed randomly in a row. a) What is the probability that the numbers 1, 2 are placed successively? b) What is the probability that the numbers $k, k+1, k+2$ are placed successively?

Exercise 11. From a deck of n cards, p cards are extracted one after the other and placed on a table in a row. a) Compute the number of possible extractions. b) For $n = 32$ and $p = 4$ compute the probability that only the second card is an ace.

Exercise 12. From the first 100 integers $1, 2, 3, \dots, 100$, we randomly choose one. What is the probability that it is a multiple of 2, 5 or 11?

Exercise 13. If n letters are randomly placed in n envelopes, what is the probability that at least one letter is placed in the corresponding envelope?

Exercise 14. 10 balls are placed in 4 boxes. What is the probability that at least one box remains empty?

Exercise 15. The coefficients a, b, c of the equation $ax^2 + bx + c = 0$ are determined by rolling three dice. What is the probability that: a) the equation has two real roots; b) the equation has two complex roots; c) the number -1 is a solution of the equation.

Exercise 16. A coin is tossed three times. What is the probability that exactly two heads occur, given that: a) the first outcome was a head? b) the first outcome was a tail? c) the first two outcomes were heads? d) the first two outcomes were tails? e) the first outcome was a head and the third outcome was a head?

Exercise 17. A die is rolled twice. What is the probability that the sum of the faces is greater than 7, given that: a) the first outcome was a 4? b) the first outcome was greater than 3? c) the first outcome was a 1? d) the first outcome was less than 5?

Exercise 18. 50 lottery tickets are sold, only one of them being the winner. Each buyer can choose from those tickets which have not been previously sold. Which player do you prefer to be? The first? The second? The third?

Exercise 19. Suppose that 9 out of 500 chips in a particular box are defective, and suppose that 3 chips are sampled at random from the box, without replacement. If each of the 3 chips sampled is tested to determine whether it is defective (1) or satisfactory (0), then the sample space has 8 outcomes. Compute the probability of each of these 8 outcomes.

Exercise 20. A system has 4 computers. Computer 1 works with a probability of 0.88, computer 2 works with a probability of 0.78, computer 3 works with a probability of 0.92 and computer 4 works with a probability of 0.85. Suppose that the operations of the computers are independent of each other. What is the probability that the system works in each of the following cases?

- (a) the system works only if all four computers are working;
- (b) the system works if at least one of the four computers works;
- (c) the system works if at least three of the four computers work.

Exercise 21. Two players A and B play a game of 32 cards. When a card is drawn from the deck, it is not re-introduced. Player A draws two cards and wins if he gets two figures (J, Q, K or A). If A does not win, player B draws a card, and wins if he gets a figure. Which player has bigger chances to win? What is the probability that the game does never stop?

Exercise 22. A box contains 10 white balls and 20 red balls. n balls are extracted successively, putting back the ball if it is white and not putting it back if it is red. What is the probability that exactly one red ball is extracted?

Exercise 23. Two players toss successively a coin, with the probability p that heads occur. Player A starts the game. Player A wins if he obtains heads, and in this case, the game ends. Otherwise, player B tosses the coin, and wins if he obtains tails, in which case, the game ends.

- 1. What is the probability that player A wins exactly at the n -th toss?
- 2. What is the probability that A wins?
- 3. What is the probability that the game does not end?
- 4. What should be the value of p such that the two players have equal chances of winning the game?

Exercise 24. Bag A contains 3 red balls and 7 blue balls. Bag B contains 8 red balls and 4 blue balls. Bag C contains 5 red balls and 11 blue balls. A bag is chosen at random, with each bag being equally likely to be chosen, and then a ball is chosen at random from that bag. Calculate the probabilities that: (a) a red ball is chosen; (b) a blue ball is chosen; (c) a red ball from bag B is chosen. If it is known that a red ball is chosen, what is the probability that it comes from bag A? If it is known that a blue ball is chosen, what is the probability that it comes from bag B?

Exercise 25. A doctor assumes that a patient has one of three diseases d_1 , d_2 , or d_3 . Before any test, he assumes an equal probability for each disease. He carries out a test that will be positive with probability 0.8 if the patient has d_1 , 0.6 if he has disease d_2 , and 0.4 if he has disease d_3 . Given that the outcome of the test was positive, what probabilities should the doctor now assign to the three possible diseases?

Exercise 26. A doctor gives a patient a test for a particular cancer. Before the results of the test, the only evidence the doctor has to go on, is that 1 man in 1000 has cancer. Experience has shown that, in 99 percent of the cases in which cancer is present, the test is positive; and in 95 percent of the cases in which it is not present, it is negative. If the test turns out to be positive, what probability should the

doctor assign to the event that cancer is present?

Exercise 27. The weather on a particular day is classified as either cold, warm or hot. There is a probability of 0.15 that it is cold and a probability of 0.25 that it is warm. In addition, on each day it may either rain or not rain. On cold days, there is a probability of 0.30 that it will rain, on warm days, there is a probability of 0.40 that it will rain and on hot days, there is a probability of 0.50 that it will rain. If it is not raining on a particular day, what is the probability that it is cold?

Exercise 28. In London, half of the days have some rain. The weather forecaster is correct $2/3$ of the time, i.e., the probability that it rains, given that she has predicted rain, and the probability that it does not rain, given that she has predicted that it won't rain, are both equal to $2/3$. When rain is forecast, Mr. Pickwick takes his umbrella. When rain is not forecast, he takes it with probability $1/3$. Find: a) the probability that Pickwick has no umbrella, given that it rains; b) the probability that it doesn't rain, given that he brings his umbrella.

2 Discrete random variables and distributions

Exercise 29. The distribution law of a discrete random variable X is:

x_i	-4	-2	1	2	3
$P(X = x_i)$	0.10	0.35	0.15	0.25	0.15

Represent graphically the distribution function.

Compute $P(X < 0)$, $P(X > -1)$, $P(-3.5 < X \leq -2)$, $P(-3.5 < X < -2)$

Give the distribution laws for the following random variables: $|X|$, $X^2 + X + 2$, $\inf(X, 1)$, $\sup(X, -X^2)$. For each of the above random variables, compute the median, the lower and the upper quartile and the modulus.

Exercise 30. Let there be $\theta \in [0, 0.5)$ and X a discrete random variable with values in $\{0, 1, 2, 3\}$, whose distribution law is given by $P(X = 0) = P(X = 3) = \theta$ and $P(X = 1) = P(X = 2) = 0.5 - \theta$. We further consider the random variables $Y = X(X - 1)(X - 2)(X - 3)$ and $Z = \frac{X(X-1)(X-2)}{6}$. Express the distribution functions and compute the median, the lower and the upper quartile and the modulus for X , Y and Z .

Exercise 31. A perfectly balanced die A has the number 1 on four faces and -2 on the other two. An unbalanced die B has on its six faces the numbers $-2, -1, 0, 1, 2, 3$. The probabilities of the appearances of the faces of the die B , are a geometric progression with common ratio $1/2$. Write the distribution laws for A and B . The two dice are rolled. We denote X the discrete random variable equal to the sum of the numbers that occur on the two dice. Write the distribution law for X .

Exercise 32. A coin is tossed until we have a head. Let X be the number of necessary throwings until this event is accomplished. Compute the expected value and the variance of X .

Exercise 33. A fair coin is tossed three times. A player wins \$1 if the first toss is a head, but loses \$1 if the first toss is a tail. Similarly, the player wins \$2 if the second toss is a head, but loses \$2 if the second toss is a tail, and wins or loses \$3 according to the result of the third toss. Let the random variable X be the total winnings after the three tosses (possibly a negative value). Find the probability distribution, the cumulative distribution function, the median, modulus, expected value and variance of X .

Exercise 34. Compute the expected value and the variance for the following distributions:

$$X : \begin{pmatrix} 897 & 898 & 900 & 903 \\ 0.2 & 0.4 & 0.3 & 0.1 \end{pmatrix} \quad X : \begin{pmatrix} 0.015 & 0.045 & 0.75 \\ 0.3 & 0.5 & 0.2 \end{pmatrix}$$

$$X : \begin{pmatrix} 5.20 & 5.30 & 5.80 & 6.00 \\ 0.4 & 0.2 & 0.2 & 0.2 \end{pmatrix} \quad X : \begin{pmatrix} 100 & 150 & 200 \\ 0.5 & 0.3 & 0.2 \end{pmatrix}$$

Exercise 35. If X and Y are independent, identically distributed random variables with the distribution

$$\begin{pmatrix} -1 & 0 & 1 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{pmatrix}$$

determine the distribution and the variance of the random variable $X + Y$.

Exercise 36. A fair coin is tossed 100 times. What does Chebyshev's Inequality tell you about the probability that the number of heads that turn up deviates from the expected number 50 by three or more standard deviations (i.e., by at least 15)?

Exercise 37. A coin is tossed n times. How large should be the number of tosses n such that the frequency of occurrences of heads is bounded by 0.4 and 0.6 with a probability larger than 0.9?

Exercise 38. A player aims 100 darts, with the probability $p = 0.75$ of hitting the target. Show that the probability that the player hits the target more times he doesn't is larger than 0.97.

Exercise 39. We consider two events A and B for which $P(A) = \frac{1}{4}$, $P(B|A) = \frac{1}{2}$, $P(A|B) = \frac{1}{4}$ and the random variables X and Y defined as follows:

$$X = \begin{cases} 1 & \text{if } A \text{ takes place} \\ 0 & \text{if } \bar{A} \text{ takes place} \end{cases} \quad Y = \begin{cases} 1 & \text{if } B \text{ takes place} \\ 0 & \text{if } \bar{B} \text{ takes place} \end{cases}$$

Show that $\rho(X, Y) = 0$.

Exercise 40. We toss two coins, each of them having the number 1 on one face and the number 2 on the other face. We consider the random variable X "the sum of the obtained numbers" and the random variable Y "the maximum of these numbers". Compute the correlation of X and Y .

Exercise 41. Two dice are rolled and we denote $X(Y)$ the sum of points from the two dice (maximum number of points; if the numbers are identical, we take their common value). Compute the correlation of X and Y .

Exercise 42. We consider a box containing 5 balls numbered from 1 to 5. We simultaneously take out three balls from the box. We denote X the smallest number obtained. Give the distribution law for the random variable X . Compute the expected value $E(X)$ and the standard deviation $\sigma(X)$. We denote Y the largest number obtained. Are the random variables X and Y independent? Compute the correlation coefficient $\rho(X, Y)$.

Exercise 43. We toss a perfectly balanced coin, we denote X_1 the obtained result: $X_1 = 0$ if we obtain the head, $X_1 = 1$ if we obtain the value. If $X_1 = 0$ we toss a tricked coin for which the probability to obtain the value is $2/3$. If $X_1 = 1$ we toss again the balanced coin. We denote X_2 the result of the second throwing ($X_2 = 0$ if we get the head and $X_2 = 1$ if we get the value). Let $X = X_2$ and $Y = X_1 + X_2$.

1. Give the distribution law for X_2
2. Find the joint distribution of the variables X and Y
3. Compute the law of Y conditioned by X .
4. Compute $E(X)$, $E(Y)$, $V(X)$, $V(Y)$, $Cov(X, Y)$, $\rho(X, Y)$.

Exercise 44. A sack contains n counters numbered from 1 to n . We successively extract 2 counters. Let X be the number of the first counter and Y the number of the second counter extracted. Compute $Cov(X, Y)$.

Exercise 45. A box contains a white balls and b black balls ($a + b \geq 3$). We successively take out balls from the box. Let X , Y and Z be discrete random variables, equal to 1 if the first, the second respectively the third ball extracted is white, and equal to 0 in the contrary. Determine the distribution of the random vector (Y, Z) . Find the distribution laws for Y and Z . Compute $Cov(Y, Z)$ and $\rho(Y, Z)$.

Exercise 46. We pick up a number X from the set $\{1, 2, 3, 4, 5, 6, 7\}$ and afterwards a number Y from the set $\{1, \dots, X\}$. Find the distribution of the vector (X, Y) , compute $\rho(X, Y)$.

Exercise 47. We roll a die N times and denote Z "the times we obtained 6". We toss Z times a tricked coin (with the probability p of obtaining the head). We denote X "the number of heads obtained" and Y "the number of values obtained". Compute:

1. the distribution, the expected value and the variance of Z
2. $P(X = k|Z = n)$ and $P((X = k) \cap (Z = n))$ for $k, n \in \mathbb{N}$
3. the distribution, the expected value and the variance of X and Y (hint: $C_n^k C_N^n = C_N^k C_{N-k}^{n-k}$)
4. $cov(X, Y)$ and $\rho(X, Y)$

Exercise 48. Alin and Barbu play a game with the following rule: A tosses two coins and B three coins; if A obtains more heads than B then he gets 10 RON, if he obtains the same number of heads as B he gets 1 RON and if he obtains less heads than B then he loses 5 RON. Who do you prefer to be? Alin or Barbu? Compute the correlation of the winnings of Alin and Barbu.

Exercise 49. In a library, there are books numbered from 1 to n . We randomly take out the books. We say we have a meeting, if the number from the book is the same as the extraction number. Compute the expected value and the variance for the total number of meetings.

Binomial distribution.

Exercise 50. What is the probability that when tossing a coin 10 times, we obtain:

a) three heads? b) at least one head? c) at most one head?

Exercise 51. The statistical data of a hospital shows that 20% of the patients suffering from a given illness die. If 7 patients have been diagnosed with this illness, what is the probability that:

a) all patients get well? b) at least 3 patients die? c) 4 patients die?

Exercise 52. On the average, only 1 person in 1000 has a particular rare blood type. (a) Find the probability that, in a city of 10,000 people, no one has this blood type. (b) How many people would have to be tested to give a probability greater than $1/2$ of finding at least one person with this blood type?

Exercise 53. Assume that the probability that there is a significant accident in a nuclear power plant during one year's time is .001. If a country has 100 nuclear plants, estimate the probability that there is at least one such accident during a given year.

Exercise 54. An archer hits a bull's-eye with the probability of 0.09 and the results of different attempts can be taken as independent of each other. If the archer shoots 9 arrows, calculate the probability that:

a) exactly two arrows score bull's-eyes; b) at least two arrows score bull's-eyes. What is the expected number of the bull's-eyes scored?

Exercise 55. A multiple choice quiz consists of ten questions each with five possible answers from which only one is correct. A student passes the quiz if seven or more of his answers are correct. What is the probability that a student who guesses blindly at all of the questions will pass the test? What is the probability that the student passes the test if at every question he can eliminate three incorrect answers and then guesses between the remaining two?

Geometric distribution. Negative binomial distribution.

Exercise 56. A die is rolled until we obtain a 6. Let X be the number of necessary rolls until this event is accomplished. Compute $E(X)$.

Exercise 57 (Waiting lines). Suppose a line of customers waits for service at a counter. It is often assumed that, in each small time unit, either 0 or 1 new customers arrive at the counter. The probability that a customer arrives is p and that no customer arrives is $q = 1 - p$. The time T until the next arrival has a geometric distribution. What is the probability that no customer arrives in the next k time units, that is $P(T > k)$? What is the probability that the customer's service takes s more time units, given that the customer has already been served for r time units?

Exercise 58. Two players A and B take part at a competition. Each turn of the competition may be won by player A with probability p or by player B with probability $(1 - p)$. The competition ends either when A wins m turns or when B wins n turns. Find the probability that player A wins the competition at turn k .

Exercise 59. Consider the same situation as in Ex. 5. (a) If the archer shoots a series of arrows, what is the probability that the first bull's-eye is scored with the fourth arrow? (b) What is the probability that the third bull's-eye is scored with the tenth arrow? (c) What is the expected number of the arrows shot before the first bull's-eye is scored? (d) What is the expected number of the arrows shot before the third bull's-eye is scored?

Hypergeometric distribution.

Exercise 60. A bridge deck has 52 cards with 13 cards in each of four suits: spades, hearts, diamonds, and clubs. A hand of 13 cards is dealt from a shuffled deck. Find the probability that the hand has (a) a distribution of suits 4, 4, 3, 2 (for example, four spades, four hearts, three diamonds, two clubs). (b) a distribution of suits 5, 3, 3, 2.

Exercise 61. A manufactured lot of buggy whips has 20 items, of which 5 are defective. A random sample of 5 items is chosen to be inspected. Find the probability that the sample contains exactly

one defective item (a) if the sampling is done with replacement. (b) if the sampling is done without replacement.

Multinomial distribution

Exercise 62. Consider the same situation as in Ex. 5. Suppose that the archer misses the target completely with a probability of 0.12. If the archer shoots eight arrows whose performances are independent of each other, what is the probability that (a) the archer scores exactly two bull's-eyes and misses the target exactly once; (b) the archer scores exactly one bull's-eye and misses the target exactly twice. What is the expected number of times the archer misses the target?

Poisson distribution.

Exercise 63. The mean density of microbes per m^3 is 100. Taking a sample of $2dm^3$ compute the probability that there is at least one microbe in this sample.

Exercise 64. A baker blends 600 raisins and 400 chocolate chips into a mix and, from this, makes 500 cookies. (a) Find the probability that a randomly picked cookie will have no raisins. (b) Find the probability that a randomly picked cookie will have exactly two chocolate chips. (c) Find the probability that a randomly chosen cookie will have at least two bits (raisins or chips) in it.

Exercise 65 (London bombing). The statistics of flying bomb hits in an area in the south of London during the Second World War provide the following data. The area in question was divided into $24 \times 24 = 576$ small areas. The total number of hits was 537. There were 229 squares with 0 hits, 211 with 1 hit, 93 with 2 hits, 35 with 3 hits, 7 with 4 hits, and 1 with 5 or more. Assuming the hits were purely random, use the Poisson approximation to find the probability that a particular square would have exactly k hits. Compute the expected number of squares that would have 0, 1, 2, 3, 4, and 5 or more hits and compare this with the observed results.

Exercise 66. Assume that, during each second, a police station receives one call with probability .01 and no calls with probability .99. Use the Poisson approximation to estimate the probability that the operator will miss at most one call if she takes a 5-minute coffee break.

Exercise 67. An average of 2 customers arrive at a post office every minute. Assuming that that number of customer arrivals has a Poisson distribution, compute: a) The average time period between two arrivals; b) The probability that at least 2 minutes elapse between two arrivals.

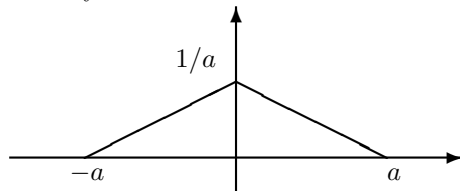
Exercise 68. Assume that the number of costumers of a shop is Poisson distributed, with the parameter λ . In this shop, the probability that the wallet of a customer is stolen is equal to p . What is the distribution of the number of stolen wallets?

3 Continuous random variables and distributions

Exercise 69. Let X be a random variable whose distribution function is $F(x) = a + b \arctan \frac{x}{2}$, $x \in \mathbb{R}$. Find the constants a and b such that F satisfies the properties of a distribution function.

Exercise 70. The density function of a continuous random variable X is $f(x) = \frac{b}{a^2 + x^2}$, $x \in \mathbb{R}$. Find the coefficient b and the distribution function of X .

Exercise 71. The graph of the density function of a continuous random variable X is shown below.



Find the density and distribution functions, the expected value, the variance, the modulus and the median of X .

Exercise 72. Consider the function:

$$f(x) = \begin{cases} 0 & , \text{ if } x < 0 \\ a \sin x & , \text{ if } x \in [0, \pi] \\ 0 & , \text{ if } x > \pi \end{cases}$$

- a) Find the coefficient a such that f is the density function of a continuous random variable X , and compute the distribution function of X .
 b) Compute $E(X)$ and $V(X)$.
 c) What is the probability that X takes values inside the interval $(0, \frac{\pi}{4})$?

Exercise 73. The same questions as in the previous problem for the function

$$f(x) = \begin{cases} ax^2 & \text{if } x \in [-1, 1] \\ 0 & \text{if } x \in (-\infty, -1) \cup (1, \infty) \end{cases}$$

considering the interval $[0.5, 2]$ at point c).

Exercise 74. Consider the function

$$f(x) = \begin{cases} k^2 x e^{-kx} & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$

- a) Verify that f is the density function of a continuous random variable X , for any $k > 0$.
 b) Find the distribution function of X , $E(X)$ and $V(X)$.
 c) Assuming that $E(X) = 20$, compute k .

Exercise 75. A number U is chosen at random in the interval $[0, 1]$. Find the probability that (a) $R = U^2 < 1/4$; (b) $S = U(1 - U) < 1/4$; (c) $T = U/(1 - U) < 1/4$. Find the distribution function F and the density function f for each of the random variables R , S , and T as well as their expected values and variances.

Exercise 76. A point P in the unit square has coordinates X and Y chosen at random in the interval $[0, 1]$. Let D be the distance from P to the nearest edge of the square, and E the distance to the nearest corner. What is the probability that (a) $D < 1/4$; (b) $E < 1/4$? Find the distribution F and density f for the random variable D .

Exercise 77. An expert witness in a paternity suit testifies that the length (in days) of a pregnancy, from conception to delivery, is approximately normally distributed, with parameters $\mu = 270$, $\sigma = 10$. The defendant in the suit is able to prove that he was out of the country during the period from 290 to 240 days before the birth of the child. What is the probability that the defendant was in the country when the child was conceived?

Exercise 78. A final examination is constructed so that the test scores are approximately normally distributed, with parameters μ and σ . The instructor assigns letter grades to the test scores as follows: if $\mu + \sigma < x$ then grade A; if $\mu < x < \mu + \sigma$ then grade B; if $\mu - \sigma < x < \mu$ then grade C; if $\mu - 2\sigma < x < \mu - \sigma$ then grade D; if $x < \mu - 2\sigma$ then grade F. This is the process of grading on the curve. What fraction of the class gets A, B, C, D, F?

Exercise 79. Bearing Works manufactures bearing shafts whose diameters are normally distributed with parameters $\mu = 1$, $\sigma = .002$. The buyers specifications require these diameters to be $1.000 \pm .003$ cm. What fraction of the manufacturers shafts are likely to be rejected? If the manufacturer improves her quality control, she can reduce the value of σ . What value of σ will ensure that no more than 1 percent of her shafts are likely to be rejected?

Exercise 80. The lifetime of an electronic tube is a continuous random variable X with exponential density $f(x) = b e^{-bx}$ $x \geq 0$. Find $P(k \leq X \leq k + 1)$ and the expected lifetime of the tube.

Exercise 81. Suppose that the time (in hours) required to repair a car is an exponentially distributed random variable with parameter $\lambda = 1/2$. What is the probability that the repair time exceeds 4 hours? If it exceeds 4 hours what is the probability that it exceeds 8 hours? What is the expected time required to repair the car?

Exercise 82. Suppose that the number of years a car will run is exponentially distributed with parameter $\lambda = 1/4$. If you buy a used car today, what is the probability that it will still run after 4 years?

Exercise 83. Suppose you are watching a radioactive source that emits particles at a rate described by an exponential density with parameter $\lambda = 1$. Find the probability that a particle will appear (a) within the next second; (b) within the next 3 seconds; (c) between 3 and 4 seconds from now; (d) after 4 seconds from now. Compute the expected time until the next particle appears.

4 Central limit theorem and normal approximation

Exercise 84. Consider rolling a die N times, the set of data representing the number which appears is 1; 2; 3; 4; 5; 6. Construct the sampling distribution of sample means, if samples of size 5 are considered. Compute the mean and the standard deviation of the sampling distribution and represent it's histogram.

Exercise 85. Calculate the following probabilities both exactly and by using a normal approximation:

- (a) $P(X \geq 8)$ where $X \sim \text{Bin}(10, 0.7)$
- (b) $P(2 \leq X \leq 7)$ where $X \sim \text{Bin}(15, 0.3)$
- (c) $P(X \leq 4)$ where $X \sim \text{Bin}(9, 0.4)$
- (d) $P(8 \leq X \leq 11)$ where $X \sim \text{Bin}(14, 0.6)$

Exercise 86. Suppose that a fair coin is tossed n times. Estimate the probability that the proportion of heads obtained lies between 0.49 and 0.51 for $n = 100, 200, 500, 1000$ and 2000.

Exercise 87. Consider a sample of normally distributed random variables $X_1, X_2, \dots, X_n$ with mean μ and variance $\sigma^2 = 7$. If $n = 15$, what is the probability that $|\mu - \bar{X}| \leq 0.4$? What is the probability if $n = 50$?

Exercise 88. The capacitances of certain electronic components have a normal distribution with a mean $\mu = 174$ and a standard deviation $\sigma = 2.8$. If an engineer randomly selects a sample of size $n = 30$ components and measures their capacitances, what is the probability that the engineer's point estimate of the mean μ will be within the interval (173, 175)?

Exercise 89. The weights of bricks are normally distributed with $\mu = 110$ and $\sigma = 0.4$. If the weights of 22 randomly selected bricks are measured, what is the probability that the resulting point estimate of μ will be in the interval (109.9, 110.1)?

Exercise 90. The pH levels of food items prepared in a certain way are normally distributed with a standard deviation $\sigma = 0.82$. An experimenter estimates the mean pH level by averaging the pH levels of a random sample of n items.

- (a) If $n = 5$, what is the probability that the experimenter's estimate is within 0.5 of the true mean value?
- (b) If $n = 10$, what is the probability that the experimenter's estimate is within 0.5 of the true mean value?
- (c) What sample size n is needed to ensure that there is a probability of at least 99% that the experimenter's estimate is within 0.5 of the true mean value?

5 Confidence intervals. Hypothesis testing.

Exercise 91. The mean length of time required for returning students to register at our university has been 90 minutes, and the standard deviation is 15 minutes. A new "faster" registration procedure was introduced this semester. The students seem to think the registration people "blew it". The students collected a sample of size $n = 45$ and found the average registration time $\bar{x} = 95.6$. Does the sample show sufficient evidence, at the 0.05 level of significance, to support their contention?

Exercise 92. A manufacturer of a certain cigarette claims that is cigarette averages no more then 17.5 milligrams of nicotine and the standard deviation is 1.8 milligrams. A sample of 42 randomly selected cigarettes was tested and the mean of 18.3 milligrams was observed. At the 0.05 level of significance, do we have sufficient evidence to reject the manufacturer's claim?

Exercise 93. Waiting times, in hours, at a popular restaurant are believed to be approximately normally distributed with a variance of 2.25 hours during busy periods. (a) A sample of 20 customers revealed a mean waiting time of 1.52 hours. Construct the 95% confidence interval for the estimated of the population mean. (b) Suppose that the mean of 1.52 hours had resulted from a sample of 32 customers. Find the 95% confidence interval. What effect does a larger sample size have on the confidence interval?

Exercise 94. 85 of the newly admitted freshmen at a college were randomly selected and asked to perform a physical fitness task. The times recorded had a mean value of 12.87 minutes. The time required to complete the task is distributed with a standard deviation of 1.3 minutes. Construct the 90% confidence interval estimate for the mean time required by all freshmen.

Exercise 95. The admissions office at a hospital recently stated that the mean age of its patients was 42 years. A random sample of 120 ages was collected from the records of the admissions office in attempt

to disprove the claim. Is a sample mean of 44.2 significantly larger than the claimed 42 years, at the 0.05 level of significance, considering $\sigma = 20$ years?

Exercise 96. A student group claims that the average student must travel for at least 25 minutes in order to reach the college every day. The college admissions office obtained a random sample of 36 one-way travel times from students. The sample had a mean of 19.4 minutes and a standard deviation of 9.6 minutes. Does the admissions office have sufficient evidence to reject the students' claim? Use $\alpha = 0.01$.

Exercise 97. Ten randomly selected shut-ins were each asked to list how many hours of television they watched per week. The results are:

82 66 90 84 75 88 80 94 110 91

Determine the 90% confidence interval estimate for the mean number of hours of television watched per week by shut-ins.

Exercise 98. It is claimed that students at a certain university will score an average of 35 on a given test. Is the claim reasonable if a random sample of test scores from the university yields:

33 42 38 37 30 42

Complete the hypothesis test using $\alpha = 0.05$.

Exercise 99. A commercial farmer harvest his entire field of a vegetable crop at one time. Therefore, he would like to plant a variety of green beans that mature all at one time (small standard deviation between maturity times of individual plants). A seed company has developed a new hybrid strain of green beans that it believes to be better for the commercial farmer. The maturity time of the standard variety has an average of 50 days and a standard deviation of 2.1 days. A random sample of 30 plants of the new hybrid showed a standard deviation of 1.65 days. Does this sample show a significant lowering of the standard deviation, at the 0.05 level of significance?

Exercise 100. Use the sample data given in Problem 8 to construct the 0.95 confidence interval estimate for the standard deviation of the test scores.

Exercise 101. Air pollution is determined by measuring several different elements that can be detected in the air. One of them is carbon monoxide. The following sample of daily readings was obtained from a local newspaper:

3.5 3.9 2.8 3.1 3.1 3.4 4.8 3.2 2.5 3.5 4.4 3.1

- Compute the mean and the standard deviation of the sample.
- Carbon monoxide is measured and interpreted according to the accompanying scale. Does the sample show sufficient evidence to allow us to conclude that the carbon level monoxide is low, that is, $\mu < 4.9$ at $\alpha = 0.05$?
- Does the sample show sufficient evidence to allow us to reject the claim that the variance in the carbon monoxide readings is no more than 0.25 at $\alpha = 0.05$?
- Construct the 90%, 95% and 99% confidence intervals for estimating the mean daily level of carbon monoxide pollution.
- Construct the 90%, 95% and 99% confidence intervals for estimating the standard deviation of carbon monoxide pollution.

Exercise 102. It has been suggested that abnormal human males tend to occur more in children born to older-than-average parents. Case histories of 20 abnormal males were obtained and the ages of 20 mothers were:

31 21 29 28 34 45 21 41 27 31 43 21 39 38 32 28 37 28 16 39

The mean age at which mothers in the general population give birth is 28 years.

- Compute the mean and the standard deviation of the sample.
- Does the sample show sufficient evidence to support the claim that abnormal male children have older-than-average mothers? Use $\alpha = 0.05$.

- c. Compute the 95% confidence interval for estimating the standard deviation of the ages of all mothers for these children.

Exercise 103. A production process is considered to be out of control if the produced parts have a mean length different from 27.5 millimeters or a standard deviation that is greater than 0.5 millimeter. A sample of 30 parts yields a sample mean of 27.63 millimeters and a sample standard deviation of 0.87 millimeters.

- At the 0.05 level of significance, does this sample indicate that the process should be adjusted in order to correct the standard deviation of the product?
- At the 0.05 level of significance, does this sample indicate that the process should be adjusted in order to correct the mean value of the product?

Exercise 104. In a large cherry orchard, the average yield has been 4.35 tons per acre for the last several years. A new fertilizer was tested on 15 randomly selected 1-acre plots. The yields from these plots follow:

3.56 5.00 4.88 4.93 3.92 4.25 5.12 5.13 4.79 4.45 5.35 4.81 3.48 4.45 4.72

At the 0.05 level of significance, does this sample show sufficient evidence to claim that there was a significant increase in production?

6 Correlation. Regression

Exercise 105. Draw the scatter diagram for the following bivariate data:

x	1	1	3	3	5	5	7	7	9	9
y	1	2	2	3	3	4	4	5	5	6

Compute the correlation coefficient and find the equation of the regression line.

Exercise 106. In a simple regression analysis with $n = 18$ data points, the values $b_1 = 0.522$ with $s_{b_1} = 0.142$ have been obtained. Test the null hypothesis $\beta_1 = 0$ against a two-tailed alternative hypothesis.

Exercise 107. For a sample of 10 students, the following bivariate data represents the distance and the duration of their travel to school.

x	1	3	5	5	7	7	8	10	10	12
y	5	10	15	20	15	25	20	25	35	35

- Determine the scatter diagram and the correlation coefficient of the sample.
- Does this sample show sufficient evidence for the positive linear correlation of the distance and the duration of travel in the case of all students ?
- Find the equation of the regression line.
- Does the slope b_1 of the regression line show sufficient evidence to claim that $\beta_1 > 0$ at a significance level $\alpha = 0.05$?

Exercise 108. The tax rate is chosen such that it affects the unemployment level. The following table shows the tax rate vs. the unemployment level for a time frame of 10 months.

x	12,27	12,34	12,31	15,81	15,67	17,75	11,56	15,71	19,91	19,99	21,11
y	5,9	5,6	5,9	5,9	6,2	7,6	7,5	7,3	7,6	7,2	8,3

- Determine the scatter diagram and the correlation coefficient of the sample.
- Does this sample show sufficient evidence for the positive linear correlation of the tax rates and unemployment levels in general ?
- Find the equation of the regression line.
- Does the slope b_1 of the regression line show sufficient evidence to claim that $\beta_1 > 0$ at a significance level $\alpha = 0.05$?